

# PATNALA UDAY KUMAR

SOFTWARE ENGINEER | FULL-STACK & BACKEND DEVELOPER

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## PROFESSIONAL SUMMARY

Software engineer focused on full-stack and backend development with hands-on experience building and deploying applications using Java, Python, JavaScript, React.js, Node.js, and FastAPI. Designs REST APIs, secure authentication, SQL/SQLite data layers, and responsive interfaces, with AI integrations using Gemini and browser-based face detection.

## TECHNICAL SKILLS

**Languages:** Java, Python, SQL, JavaScript

**Frontend:** React.js, HTML5, CSS3, Tailwind CSS, Responsive Design

**Backend & APIs:** Node.js, FastAPI, REST APIs, API Integration

**Core & Tools:** Object-Oriented Programming, Data Structures, Git, GitHub, Vercel

## PROJECTS

### Music Mirror | React.js, Python, FastAPI, face-api.js

GitHub: [github.com/UdayPatnala/music-mirror](https://github.com/UdayPatnala/music-mirror) | Live Demo: <https://music-mirror-aos.vercel.app/>

- Developed a real-time webcam face-detection pipeline in React.js and face-api.js that calls the FastAPI POST /recommend endpoint to normalize expressions and return curated tracks.
- Integrated embedded YouTube playback, manual mood overrides, and localStorage-backed profiles, mood history, and favorites in a responsive application deployed on Vercel.

### Churn Prediction System | Python, XGBoost, Scikit-learn, FastAPI, Docker, Streamlit

GitHub: [github.com/UdayPatnala/Churn-Prediction-System](https://github.com/UdayPatnala/Churn-Prediction-System) | Live Demo: [nebula-nmo.vercel.app](https://nebula-nmo.vercel.app)

- Engineered an end-to-end ML pipeline with Scikit-learn and XGBoost utilizing GridSearchCV hyperparameter optimization, standard scaling, and one-hot encoding for real-time customer churn prediction.
- Developed a FastAPI REST API with Pydantic payload validation and an interactive Streamlit analytics dashboard; containerized API and UI microservices using Docker Compose.

### JavaPath Pro | React.js, Node.js, Express.js, SQLite, JWT

GitHub: [github.com/UdayPatnala/Java-Path](https://github.com/UdayPatnala/Java-Path) | Live Demo: <https://javapath-pro-aos.vercel.app/>

- Built an interactive Java learning platform with Java 17 execution through the JDoodle API, real-time syntax diagnostics, and a Gemini-powered AI mentor.
- Implemented bcrypt password hashing, JWT authentication, protected REST APIs, and SQLite/Sequelize persistence for user progress and chat history; deployed the frontend on Vercel.

## INTERNSHIP EXPERIENCE

### Machine Learning Intern | Codec Technologies Pvt. Ltd. | Jun 2025 - Aug 2025

- Developed data-preprocessing and baseline machine-learning workflows using Python, NumPy, Pandas, and scikit-learn during an 8-week internship.

### Web Developer Intern | Codec Technologies Pvt. Ltd. | Jan 2026 - Feb 2026

- Built and tested responsive web interfaces using HTML, CSS, JavaScript, DOM manipulation, and debugging during a 4-week internship.

## EDUCATION

### B.Tech in Computer Science Engineering (Data Science) | 2022 - 2026

Raghu Institute of Technology (Autonomous), Visakhapatnam, Andhra Pradesh | **CGPA: 7.70/10**

## CERTIFICATIONS

[Programming in Java - NPTEL \(12 weeks, 2024\)](#) | [AWS Academy Cloud Foundations \(2024\)](#) | [Software Testing Masterclass - Udemy](#)

## ACHIEVEMENT

Published "Smart Music Recommendation System Based on User Emotions," IJARESM, Vol. 14, Issue 3 (Mar 2026).